# What drives a customer booking?

**Model performance**

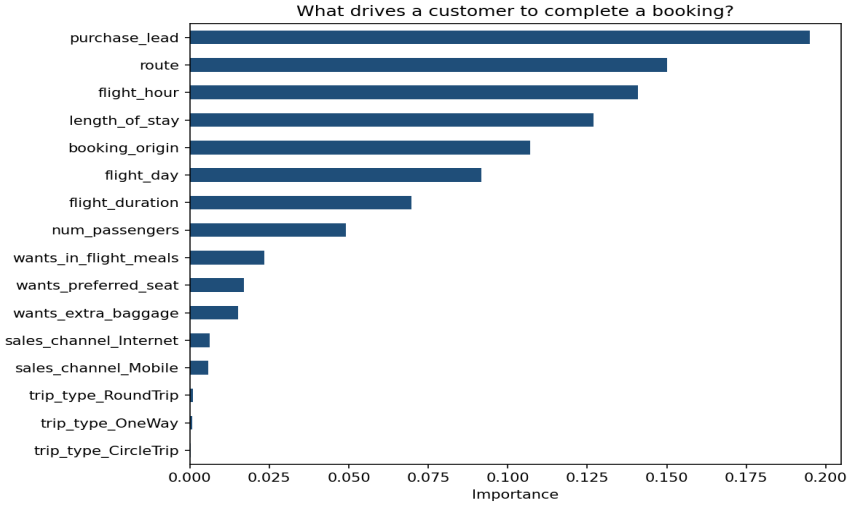
- Random Forest trained on 50,000 past customer bookings
- Tested 5 times (cross-validation): AUC 0.77, so it ranks customers well and is clearly better than guessing
- Accuracy 85%, but it only catches about 10% of real bookers, so it is best used to rank customers, not as a yes/no answer

**What matters most**

- How far ahead customers book is the #1 factor. People booking within a week book most often (17%)
- Route, flight time, length of stay and country also matter
- Sales channel and trip type have almost no effect

**Recommendations**

- Use the model to rank customers and send offers to the most likely bookers first
- Focus on customers looking at flights 1 week to 3 months away
- Prioritise the biggest markets: Malaysia and the Auckland to Kuala Lumpur route




Prepared by Ahmed Khalil